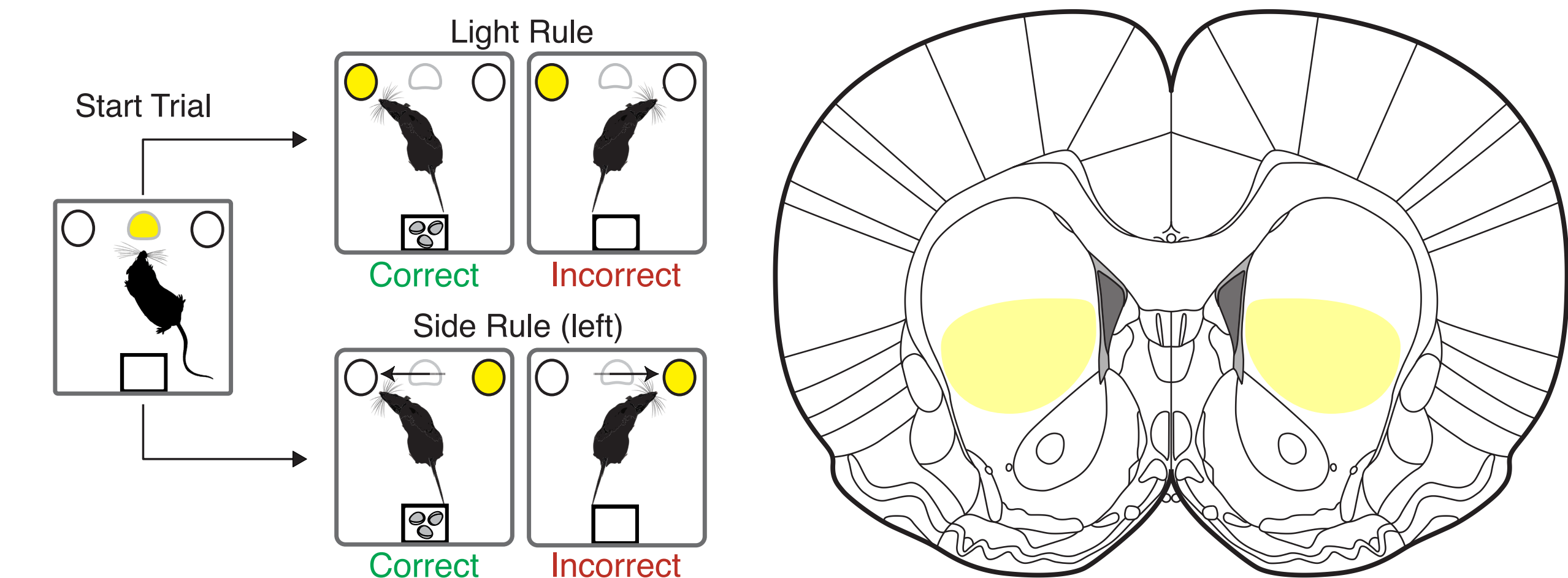


Intro

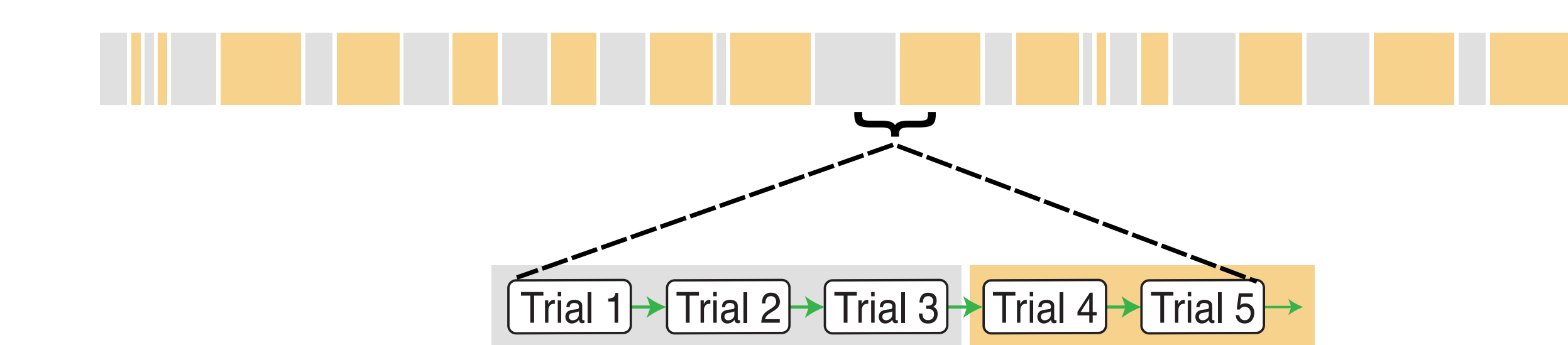
- Cognitive control, the balance between flexible and rigid thinking, is disrupted in psychiatric disorders
- The extradimensional set-shifting task has been used to probe this balance in rats and humans
- Previous work has demonstrated that electrical stimulation can enhance cognitive control for both rats and humans by decreasing reaction times in this task
- This decrease could be useful as a control signal for identifying optimal therapeutic settings but it would need to be measurable acutely in a robust, parameter-dependent manner

Methods



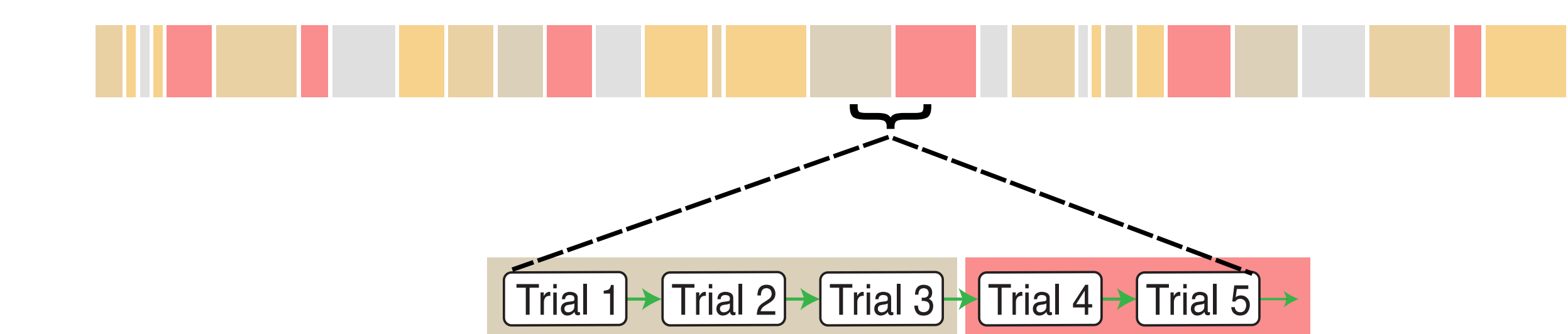
- Long-Evans rats completed either of two variants of an extradimensional set-shifting task while receiving active or sham stimulation in the mid-striatum

Block-based (BB) Group

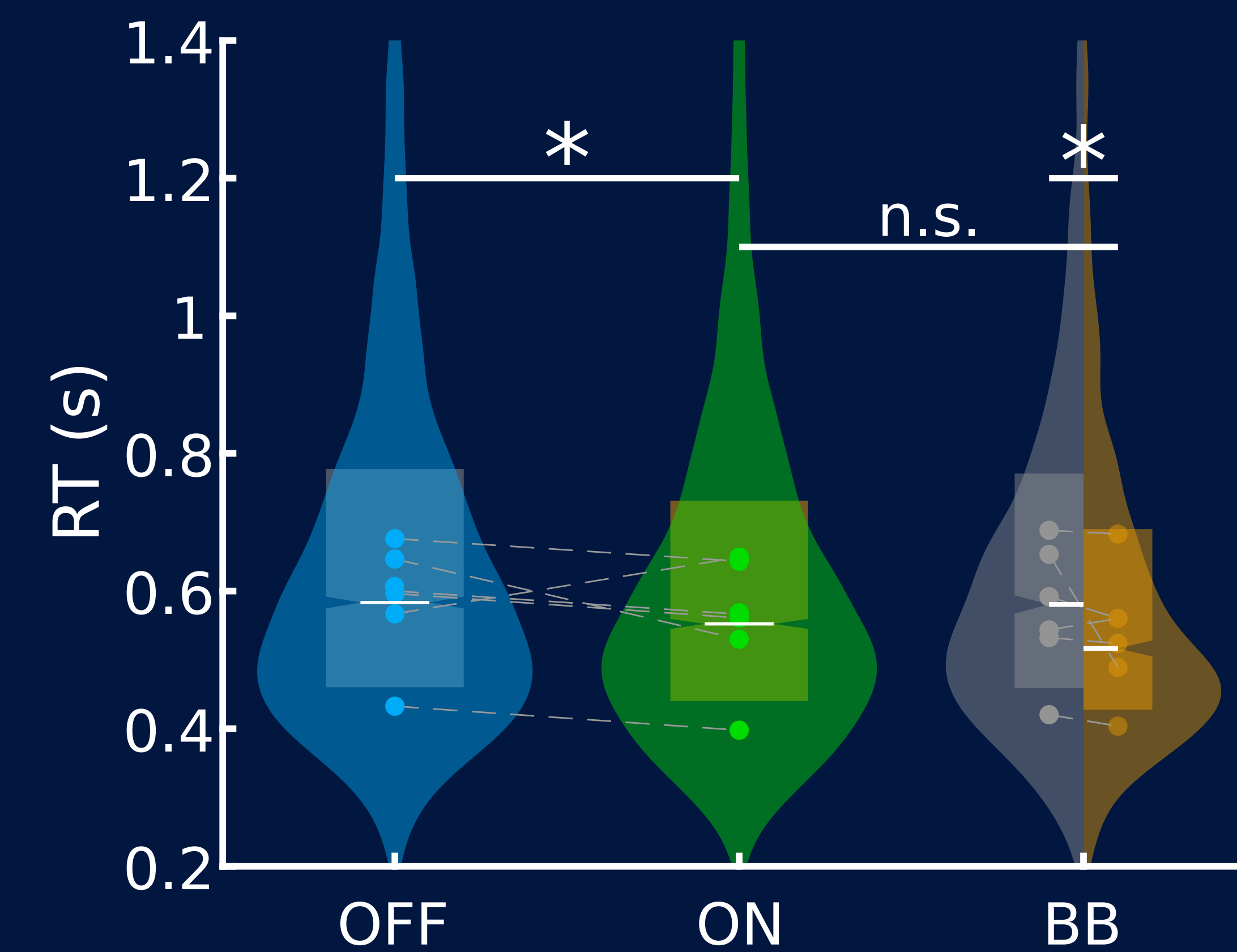
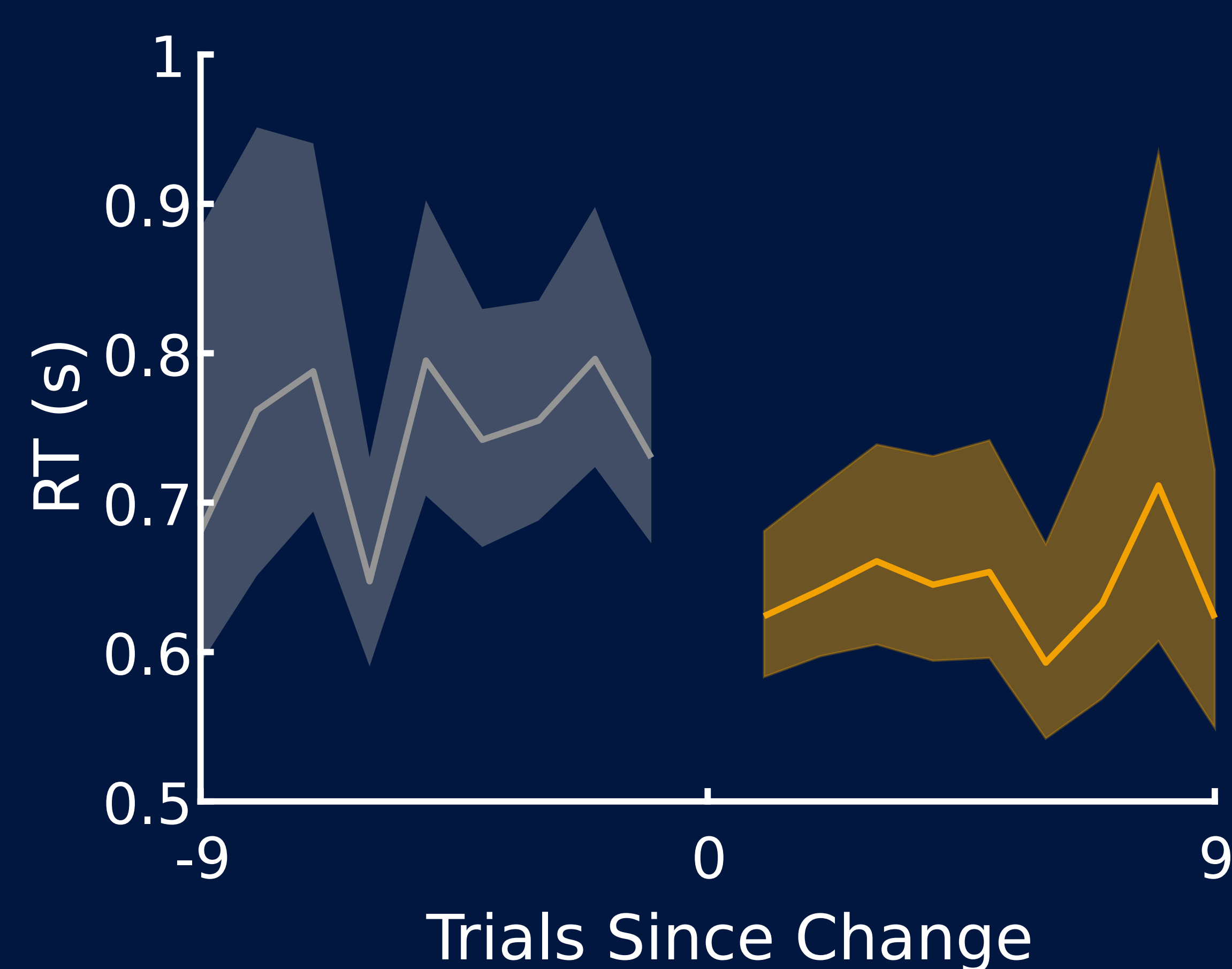


- 15 total sessions where stimulation was connected but off (sham), continuously active at 300 μ A and 130 Hz, or alternating between variable-length blocks of active and sham settings

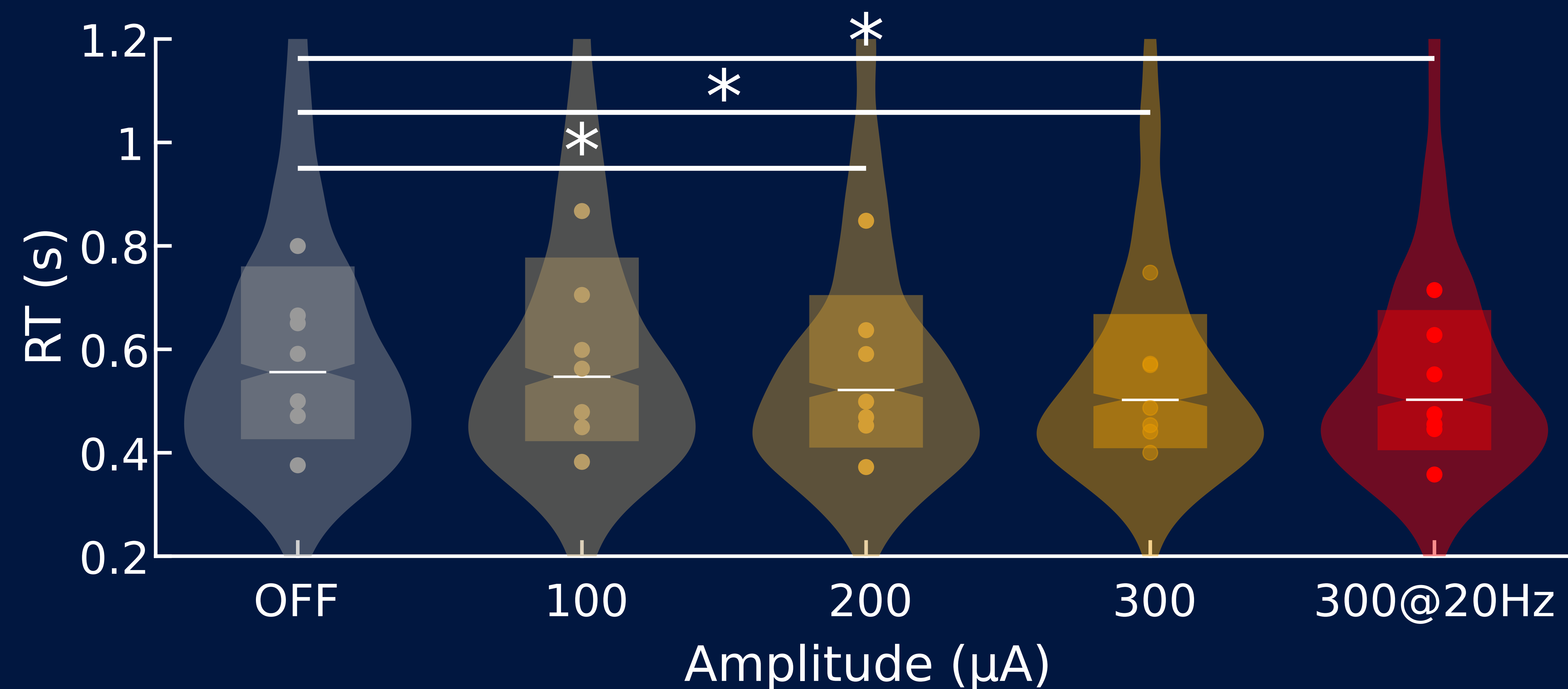
Variable-parameter (VP) Group



- 5 total sessions composed of interleaved blocks of sham stimulation, 100-300 μ A stimulation at 130 Hz, and 300 μ A stimulation at 20 Hz



Acute stimulation produced an immediate and lasting reduction in reaction time comparable to continuous stimulation.

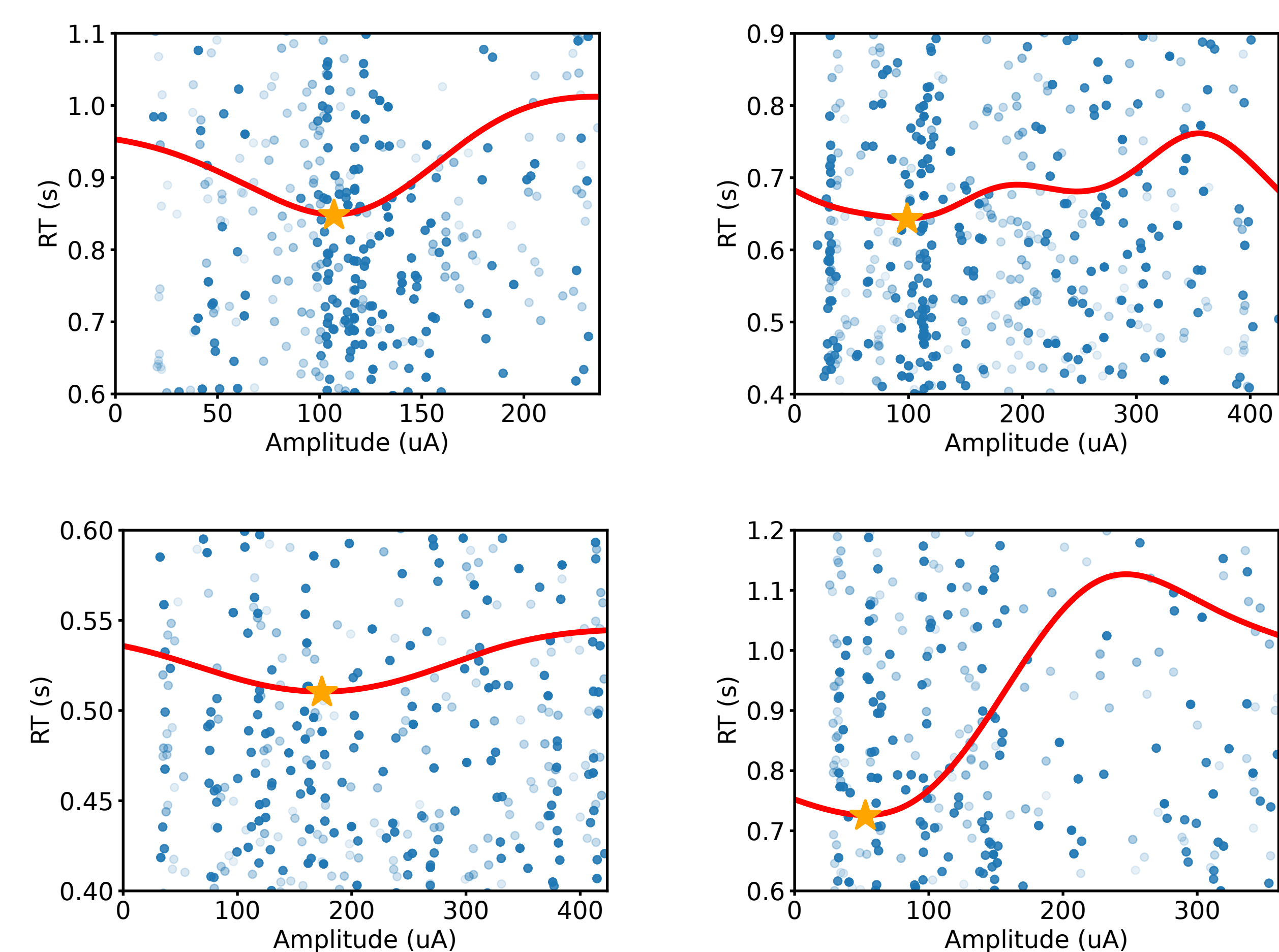


This reduction was sensitive to the choice of parameters supporting its utility as a therapeutic control signal.

Bayesian Optimization



Optimization Pilot



- Bayesian optimization of stimulation amplitude was piloted in 4 rats for 5 sessions
- The upper limit on amplitude was individually selected for each rat to ensure tolerability
- Optimization identified locations that minimized reaction time in all 4 rats

Discussion

- Acute stimulation fully replicated the cognitive control enhancements previously observed for continuous stimulation
- These enhancements were sensitive to the stimulation dose
- In a pilot study, it was possible to identify the optimal acute dose in individual rats
- In the next phase of the experiment, we will determine if the optimal acute dose translates to the continuous setting validating its utility for future therapeutic control

Acknowledgements

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